

PROBLEM SET 3—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions: Due on 22/11/2020; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

This is a very short problem set that covers one of the models from lecture 7.

1. Consider the model we saw in lecture 7 about credit rationing. For completeness sake, I write here again the main description. Consider a population of risk-neutral borrowers who each own a project that requires an initial outlay of $I = 1$ and yields a random return X , where $X \in \{0, R\}$. Let $p \in [0, 1]$ denote the probability that $X = R$. Borrowers have no wealth and must obtain investment funds from a bank. A borrower can be of two different types $i = s, r$. The borrower of type i has project with return characteristics (p_i, R_i) . Assume the following.

A1' $p_r R_r < 1 < p_s R_s$.

A2 $p_s > p_r$ and $R_s < R_r$.

As you can see, assumption A1' is different from the one we covered in class. Assume that there is excess demand for funds for the bank, i.e. $\alpha < 1$ and that available funds are sufficient to avoid crowding out either type of borrower completely, i.e. $\max\{\beta, 1 - \beta\} < \alpha$.

- (a) Compute the optimal single fixed repayment scheme D for the bank. Is there any type of borrower that is excluded for sure?
- (b) Now consider that the bank can offer a type-dependent contract that specifies the probability of financing, x_i and the repayment amount D_i if the project is successful. That is, the bank can offer a plan (x_i, D_i) for $i = s, r$. Compute the optimal contracts for the bank under assumptions A1' and A2. Show that in contrast with what we saw in class, now credit rationing reappears under two possible forms:
- If $R_s[\beta p_s + (1 - \beta)p_r] \geq 1$ then both types of project are funded and there is cross-subsidization between types. Conclude that this happens because the banks cannot turn away risky borrowers without restricting access to credit for the safe borrowers.
 - If $R_s[\beta p_s + (1 - \beta)p_r] < 1$ then neither project is funded, which is a severe form of credit rationing, i.e. *total financial collapse*.

Solution

- (a) Let D be a single-fixed repayment scheme. As seen in the lecture notes, type- i borrowers apply for funds if and only if $D \leq R_i$. If $D > R_s$ only the risky borrowers will consider applying. The maximum then that the bank can charge is $D = R_r$. In this case, the bank is only contracting with risky borrowers and obtains $(1 - \beta)(p_r R_r - 1)$. In contrast with the case analyzed in class, assumption A1' implies that this term is negative. Thus, contracting with risky borrowers only is not optimal. This implies that $D \leq R_s$, i.e. the bank should contract with both types, assuming of course that this is profitable. If this is the case, the bank should charge the highest possible repayment that the safe borrowers can afford, that is $D = R_s$. In this case, the profits obtained by the bank are given by

$$\alpha(\beta(p_s R_s - 1) + (1 - \beta)(p_r R_s - 1)).$$

In order for the bank to be willing to contract with both borrowers, the expression above must be positive. Thus, profits are positive if and only if $R_s(\beta p_s + (1 - \beta)p_r) \geq 1$ and in this case the bank finds optimal to contract with both types. On the other hand, if $R_s(\beta p_s + (1 - \beta)p_r) < 1$ the bank abstains from lending to any of the borrowers. Note that in both situations, there is credit rationing.

- (b) We need to characterize the second-best contract for the bank. A contract is defined by (x_i, D_i) where

x_i is the probability of obtaining financing and D_i is the amount to be repaid. The bank's problem is:

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (1)$$

$$\text{subject to } 0 \leq x_i \leq 1, \quad \forall i = s, r \quad (2)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (3)$$

$$x_i p_i (R_i - D_i) \geq 0 \quad \forall i = s, r \quad (4)$$

$$x_i p_i (R_i - D_i) \geq x_j p_i (R_i - D_j) \quad \forall i, j = s, r \quad (5)$$

Because of assumption A2, the risky type still has the highest willingness to repay, so it is still considered the "high type". Therefore, the previous problem is equivalent to the reduced problem:

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (6)$$

$$\text{subject to } x_s p_s (R_s - D_s) \geq 0 \quad (7)$$

$$x_r p_r (R_r - D_r) \geq x_s p_r (R_r - D_s) \quad (8)$$

$$x_r \geq x_s \quad (9)$$

$$0 \leq x_i \leq 1 \quad i = s, r \quad (10)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (11)$$

Furthermore, we know that the participation constraint for the safe type and the incentive constraint for the risky type will be binding. Solving for D_s and D_r using these binding constraints and plugging the resultant expressions into the objective yields the following problem:

$$\max_{x_s, x_r} x_s (\beta (p_s R_s - 1) - (1 - \beta) p_r (R_r - R_s)) + x_r ((1 - \beta) (p_r R_r - 1)) \quad (12)$$

$$\text{subject to } 0 \leq x_s \leq x_r \leq 1 \quad (13)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (14)$$

Notice that in the objective (12), the term attached to x_r is strictly negative. Thus the bank would like to shutdown any interaction with the risky types. However, if $x_r = 0$ then it would force x_s to be zero also because of the monotonicity constraint. Since it might still be profitable to contract with the safe type, then $x_s = 0$ might not always be optimal. This implies that the monotonicity constraint will be binding. To see this, suppose towards a contradiction that $\hat{x}_s$ and $\hat{x}_r$ are optimal solutions such that $\hat{x}_r = \hat{x}_s + \epsilon$ where $\epsilon > 0$. The expected profit then is

$$\hat{x}_s [R_s (\beta p_s + (1 - \beta) p_r) - 1] + \epsilon (1 - \beta) (p_r R_r - 1).$$

However, by taking instead $\tilde{x}_r = \hat{x}_s$ we have that the profit from the contract $(\hat{x}_s, \tilde{x}_r)$ is such that

$$\hat{x}_s [R_s (\beta p_s + (1 - \beta) p_r) - 1] + \epsilon (1 - \beta) (p_r R_r - 1) < \hat{x}_s [R_s (\beta p_s + (1 - \beta) p_r) - 1]$$

which contradicts the optimality of $\hat{x}_r$. Thus, the monotonicity constraint must be binding.

Let $x = x_r = x_s$. Then the problem reduces to

$$\max_{x \in [0, \alpha]} x [R_s (\beta p_s + (1 - \beta) p_r) - 1] \quad (15)$$

From the previous problem it is easy to see that we have the following optimal solution

$$x^* = \begin{cases} \alpha & \text{if } R_s (\beta p_s + (1 - \beta) p_r) \geq 1, \\ 0 & \text{if } R_s (\beta p_s + (1 - \beta) p_r) < 1. \end{cases}$$

With this solution, it is easy to find the optimal repayment scheme from the binding incentive and participation constraints: $D_r = D_s = D^* = R_s$. Thus, the optimal contract is a *pooling contract* and it

is the same as the one found in the previous item. Furthermore, the solution implies that if the bank decides to actually fund any of the consumers, it is cross-subsidizing the risky types with the profits obtained from the safe ones. If this is the case, there is still credit rationing because the probability of funding any type of consumer is α . When the bank decides not to contract with any type, there is a total financial collapse, since no consumer gets credit.